

The Mixed-Base Tetration Phase Map Is a Circle Diffeomorphism

Paper II of the mixed-base tetration series: companion to “A Phase Law for Mixed-Base Tetration” [1]

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Abstract

Proposition 6.6 of [1] derives the exact inverse mean antisymmetry $\mu_{b,d} = -\mu_{d,b}$ for the mixed-base tetration phase under the *assumption* that $H(\theta) = \theta + \Phi_{b,d}(\theta)$ is an orientation-preserving C^1 circle diffeomorphism. This paper removes that assumption in four graded steps. **(A)** Given the phase law of [1] for the two ordered pairs (b,d) and (d,b) , the lift $\tilde{H}_{b,d} = \text{id} + \Phi_{b,d}$ is unconditionally an orientation-preserving circle *homeomorphism* with inverse $\tilde{H}_{d,b}$, and the mixed-base phase maps form a groupoid in $\text{Homeo}_+(S^1)$. **(B)** Already at this topological level, $\mu_{b,d} = -\mu_{d,b}$ holds exactly, by a Riemann–Stieltjes substitution—no smoothness whatsoever. **(C)** A minimal-regularity principle: if *both* phases are C^1 on the circle, the phase maps are automatically mutually inverse C^1 diffeomorphisms; no positivity estimate is needed, and an explicit example shows that one-sided C^1 regularity would not suffice. **(D)** Under an explicit C^1 strengthening of the channel hypotheses of [1]—differentiable strict monotonicity of the branches and quantitative positivity of the logarithmic channel, both satisfied by the regular Kneser/Paulsen–Cowgill branch—the forward phase alone is C^1 with $\tilde{H}' > 0$. The engine of (D) is an *exact* product identity for the derivative of the logarithmic tails,

$$D'_n(\theta) = \alpha T'_d(\theta) \prod_{j=1}^{n-1} \frac{\alpha T_d(j + \theta)}{\exp_b^{\circ j}(D_n(\theta))}, \quad \alpha = \log_b d,$$

in which all tower-sized factors cancel; the summability of reciprocal tower minima that drives the limit is itself automatic, since taking infima in the tower recursion yields $s_{j+1} = d^{s_j}$. In the limit,

$$\tilde{H}'_{b,d}(\theta) = \frac{\alpha T'_d(\theta)}{T'_b(\tilde{H}_{b,d}(\theta))} \prod_{j=1}^{\infty} \frac{\alpha T_d(j + \theta)}{T_b(j + \tilde{H}_{b,d}(\theta))} > 0.$$

All statements are validated numerically, including an end-to-end check on explicit C^1 tetration branches.

1 Introduction and statement of results

We keep the notation of [1]. For admissible bases a , T_a denotes the regular tetration branch with $T_a(0) = 1$ and $T_a(x+1) = a^{T_a(x)}$, and $A_a = T_a^{-1} = \text{slog}_a$ its Abel inverse. For an ordered pair (b,d) one sets $F_{b,d} = A_b \circ T_d$, $\alpha = \log_b d$, $L_b = \log_b$, and, for $x = n + \theta$ with $n \in \mathbb{N}$, $\theta \in [0, 1]$,

$$D_n(\theta) = D_n^{b,d}(\theta) := L_b^{\circ n}(T_d(n + \theta)).$$

Under Hypotheses 2.2 and 2.4 of [1], Theorems 5.6 and 5.7 of [1] give the uniform limit $D = D_{b,d} = \lim_n D_n$ on $[0, 1]$ and the phase law

$$\text{slog}_b(T_d(x)) - x = \Phi_{b,d}(\{x\}) + o(1), \quad x \rightarrow +\infty, \quad \Phi_{b,d}(\theta) = A_b(D(\theta)) - \theta, \quad (1)$$

with $\Phi_{b,d}$ continuous and, by Lemma 5.8 of [1], well defined on the circle. Write $\mu_{b,d} = \int_0^1 \Phi_{b,d}$ and $P_{b,d} = \Phi_{b,d} - \mu_{b,d}$.

Proposition 6.6 of [1] proves the exact inverse antisymmetry $\mu_{b,d} = -\mu_{d,b}$ under the standing *assumption* that

$$H(\theta) = \theta + \Phi_{b,d}(\theta)$$

is an orientation-preserving C^1 circle diffeomorphism in the lifted sense $H(\theta + 1) = H(\theta) + 1$, $H' > 0$. The purpose of this paper is to prove that assumption. Throughout, for a 1-periodic continuous $\Phi : \mathbb{R} \rightarrow \mathbb{R}$ we write

$$\tilde{H}(x) := x + \Phi(x), \quad x \in \mathbb{R},$$

for the associated *lift*; $\tilde{H}(x + 1) = \tilde{H}(x) + 1$ holds automatically. Our results form a ladder of four statements, graded by the regularity they consume: a topological theorem, a measure-level theorem, a formal C^1 principle, and an analytic C^1 theorem.

Theorem A (Homeomorphism; no smoothness required). *Assume the phase-law package (P) of Section 2 holds for both ordered pairs (b, d) and (d, b) (in particular, this is the case under Hypotheses 2.2 and 2.4 of [1] for both pairs). Then $\tilde{H}_{b,d} = \text{id} + \Phi_{b,d}$ is a strictly increasing homeomorphism of $\mathbb{R}$ commuting with the unit translation, and*

$$\tilde{H}_{b,d}^{-1} = \tilde{H}_{d,b}.$$

Consequently $\tilde{H}_{b,d}$ descends to an orientation-preserving homeomorphism $H_{b,d}$ of the circle $S^1 = \mathbb{R}/\mathbb{Z}$ with $H_{b,d}^{-1} = H_{d,b}$. More generally, whenever the package (P) holds for the pairs (p, q) , (q, r) , (p, r) , the lifts compose exactly, $\tilde{H}_{p,r} = \tilde{H}_{p,q} \circ \tilde{H}_{q,r}$; the mixed-base phase maps form a groupoid inside $\text{Homeo}_+(S^1)$.

Theorem B (Exact inverse mean antisymmetry without smoothness). *Under the hypotheses of Theorem A,*

$$\mu_{b,d} = -\mu_{d,b}$$

holds exactly. In particular, the C^1 diffeomorphism assumption in Proposition 6.6 of [1] can be dropped entirely: the conclusion of that proposition is valid whenever the phase law holds for both ordered pairs.

Theorem B explains why the inverse-pair sums in Table 1 of [1] vanish to 10^{-9} – 10^{-11} : they are exactly zero in the limit, and the observed residuals are numerical error.

Theorem C (Minimal-regularity principle). *Let $\Phi, \Psi : \mathbb{R} \rightarrow \mathbb{R}$ be 1-periodic functions of class C^1 satisfying the inverse phase equations*

$$\Psi(x + \Phi(x)) = -\Phi(x), \quad \Phi(u + \Psi(u)) = -\Psi(u), \quad x, u \in \mathbb{R}. \quad (2)$$

Then $\tilde{H} = \text{id} + \Phi$ and $\tilde{K} = \text{id} + \Psi$ satisfy $\tilde{K} \circ \tilde{H} = \tilde{H} \circ \tilde{K} = \text{id}$ and $\tilde{H}' > 0$, $\tilde{K}' > 0$ everywhere; they are lifts of mutually inverse orientation-preserving C^1 circle diffeomorphisms. Under (H1)+(H2) of Section 2, the equations (2) hold automatically for $\Phi = \Phi_{b,d}$, $\Psi = \Phi_{d,b}$ (Lemma 3.1), so two-sided C^1 regularity of the phases already implies the full diffeomorphism conclusion, with no separate positivity assumption. One-sided C^1 regularity does not suffice (Example 5.1).

Theorem D (Analytic C^1 theorem). *Assume, for the ordered pair (b, d) , Hypotheses 2.2 and 2.4 of [1] together with the strengthenings (S) and (R') of Section 2. Then:*

- (i) $D_n \rightarrow D$ in $C^1([0, 1])$, $D \in C^1([0, 1])$, and $D' > 0$ with the absolutely and uniformly convergent product representation

$$D'(\theta) = \alpha T'_d(\theta) \prod_{j=1}^{\infty} \frac{\alpha T_d(j + \theta)}{E_j(\theta)}, \quad E_j(\theta) := \exp_b^{\circ j}(D(\theta)), \quad (3)$$

where $\exp_b(t) = b^t$.

- (ii) $\Phi_{b,d}$ extends to a 1-periodic C^1 function on $\mathbb{R}$, and the lift $\tilde{H}_{b,d} = \text{id} + \Phi_{b,d}$ satisfies $\tilde{H}'_{b,d} = (A'_b \circ D) D' > 0$; explicitly,

$$\tilde{H}'_{b,d}(\theta) = \frac{\alpha T'_d(\theta)}{T'_b(\tilde{H}_{b,d}(\theta))} \prod_{j=1}^{\infty} \frac{\alpha T_d(j + \theta)}{T_b(j + \tilde{H}_{b,d}(\theta))} > 0. \quad (4)$$

- (iii) $H_{b,d}$ is an orientation-preserving C^1 circle diffeomorphism; in particular the standing assumption of Proposition 6.6 of [1] is a theorem under these hypotheses. If in addition the package (P) holds for (d, b) , then $H_{d,b} = H_{b,d}^{-1}$ is automatically C^1 as well, i.e. $\Phi_{d,b} \in C^1(S^1)$ comes for free.
- (iv) The residual converges in the C^1 topology: with $R_n(\theta) := F_{b,d}(n + \theta) - (n + \theta)$ one has $R_n \rightarrow \Phi_{b,d}$ in $C^1([0, 1])$; equivalently, $F'_{b,d}(n + \theta) \rightarrow \tilde{H}'_{b,d}(\theta)$ uniformly on $[0, 1]$.

The engine of Theorem D is the exact identity of Lemma 6.1 below, in which the mismatch constant α cancels against the first logarithmic peel and *all* tower-sized quantities cancel in closed form, leaving a product whose factors converge to 1 at the hyper-fast rate $O(1/s_j)$, $s_j = \inf_{\theta} T_d(j + \theta)$. The summability of $1/s_j$ that drives the limit is not an extra hypothesis: taking infima over the phase in the tower recursion gives $s_{j+1} = d^{s_j}$ exactly (Lemma 2.5), so the qualitative divergence $s_j \rightarrow \infty$ of [1, Hyp. 2.4] self-improves to geometric growth. As a by-product one obtains the sharp value-level comparison

$$T_b(j + \tilde{H}_{b,d}(\theta)) = \alpha T_d(j + \theta)(1 + O(1/s_j))$$

(Remark 7.1): the base-mismatch factor α sits exactly one tower level down.

The four statements interlock as follows. Theorem D, applied to the single pair (b, d) , already yields the diffeomorphism property including positivity. Theorem C shows that once *both* phases are known to be C^1 —by Theorem D applied to both pairs, or by any future route such as strip analyticity (Remark 7.3)—positivity is automatic and free of channel estimates. Example 5.1 delimits Theorem C: C^1 regularity of the forward phase alone is compatible with a critical point, so the analytic positivity of Theorem D is genuinely needed in the one-sided situation. Theorems A and B need no smoothness at all.

2 Setting and hypotheses

2.1 Branch frame

Throughout, $b, d > 1$. We use the following normalization of the regular branches, satisfied by the Kneser/Paulsen–Cowgill tetration for real bases above $e^{1/e}$ [2, 3].

(B1) $T_b : (-2, \infty) \rightarrow \mathbb{R}$ is a continuous, strictly increasing bijection onto $\mathbb{R}$ with $T_b(0) = 1$ and $T_b(x+1) = b^{T_b(x)}$ for $x > -2$. Its inverse $A_b = \text{slog}_b : \mathbb{R} \rightarrow (-2, \infty)$ then satisfies the *global Abel equation*

$$A_b(b^y) = A_b(y) + 1 \quad \text{for all } y \in \mathbb{R}. \quad (5)$$

(B2) $T_d : [0, \infty) \rightarrow (0, \infty)$ is continuous with $T_d(0) = 1$ and $T_d(x+1) = d^{T_d(x)}$ for $x \geq 0$.

Equation (5) follows from (B1): if $x = A_b(y)$ then $b^y = b^{T_b(x)} = T_b(x+1)$, so $A_b(b^y) = x+1$. Note also $T_b(-1) = 0$ and $T_b > 0$ exactly on $(-1, \infty)$, directly from the functional equation.

2.2 The phase-law package

For an ordered pair (p, q) of admissible bases we say that the *phase-law package* (P) holds if the conclusions of Theorems 5.6, 5.7 and Lemma 5.8 of [1] are available for that pair, namely: $D_n^{p,q} \rightarrow D_{p,q}$ uniformly on $[0, 1]$; the residual satisfies (1) with the $o(1)$ uniform in the phase; and $\Phi_{p,q}$ is continuous on $[0, 1]$ with $\Phi_{p,q}(1) = \Phi_{p,q}(0)$, so that its 1-periodic extension (still denoted $\Phi_{p,q}$) is continuous on $\mathbb{R}$. By [1] this package holds under Hypotheses 2.2 and 2.4 of [1] for the pair (p, q) ; the endpoint identity is re-proved in sharpened form in Lemma 2.7 below. For the diagonal pairs the package is trivial: $F_{p,p} = \text{id}$ exactly, so $\Phi_{p,p} \equiv 0$.

We label:

(H1) the package (P) holds for (b, d) ;

(H2) the package (P) holds for (d, b) .

2.3 C^1 strengthenings

The following two conditions upgrade the channel hypotheses of [1] to the C^1 category. Recall $s_j := \inf_{0 \leq \theta \leq 1} T_d(j + \theta)$ from Hypothesis 2.4 of [1].

(S) (*Smooth strictly increasing branches.*) $T_b \in C^1((-2, \infty))$ with $T'_b > 0$, and $T_d \in C^1([0, \infty))$ with $T'_d > 0$. Consequently $A_b \in C^1(\mathbb{R})$ with

$$A'_b(y) = \frac{1}{T'_b(A_b(y))} > 0, \quad y \in \mathbb{R}. \quad (6)$$

(R') (*Quantitative real channel.*) There is $N_0 \in \mathbb{N}$ such that for all $n \geq N_0$ and all $\theta \in [0, 1]$ the iterates

$$W_k^{(n)}(\theta) := L_b^{\circ k}(T_d(n + \theta)), \quad 0 \leq k \leq n,$$

are defined and satisfy $W_k^{(n)}(\theta) > 0$ for $0 \leq k \leq n-1$.

Remark 2.1. (R') is precisely the quantitative reading of (R2) in Hypothesis 2.2 of [1]: positivity of $W_0, \dots, W_{n-1}$ is what it means for the n real logarithms defining D_n to exist; the final value $W_n = D_n$ may well be negative (and is, e.g., for $(b, d) = (e, 2)$, where $D(0) = T_e(\Phi_{e,2}(0)) < 0$). See also Remarks 2.3 and 2.4.

Lemma 2.2 (Positivity propagation). *Let $T \in C^1$ satisfy the tetration equation $T(x+1) = a^{T(x)}$ ($a > 1$) on its domain as in (B1) or (B2). If $T' > 0$ on the single period $[0, 1]$, then $T' > 0$ on the whole domain (all of $(-2, \infty)$ in case (B1), all of $[0, \infty)$ in case (B2)).*

Proof. Differentiating the functional equation,

$$T'(x+1) = (\ln a) T(x+1) T'(x). \quad (7)$$

Forward: for $x \geq 0$ one has $T(x+1) > 0$, so $T' > 0$ on $[k, k+1]$ implies $T' > 0$ on $[k+1, k+2]$; induction covers $[0, \infty)$. Backward (case (B1)): for $x \in [-1, 0)$ we have $x+1 \in [0, 1)$, hence $T(x+1) > 0$ and $T'(x+1) > 0$, so (7) gives $T'(x) = T'(x+1)/((\ln a)T(x+1)) > 0$; the same step with $x \in (-2, -1)$ (where $x+1 \in (-1, 0)$, $T(x+1) > 0$) finishes the proof. $\square$

Thus (S) only needs to be checked on one fundamental interval.

Remark 2.3. For the regular Kneser/Paulsen–Cowgill branch with real bases $b, d > e^{1/e}$ [2, 3, 4], T_a is real-analytic and strictly increasing on $(-2, \infty)$ with $T_a((-\infty, \infty)) = \mathbb{R}$, so (B1)–(B2) hold. Concerning (S): we are not aware of an explicit statement $T'_a > 0$ in the literature; it is numerically evident, it is presupposed by the numerical scheme of [1, App. A] on its inversion window, and by Lemma 2.2 it only needs to be checked on a single period. (By (7), the zero set of T'_a is moreover invariant under $x \mapsto x+1$ wherever $T_a > 0$, so a single critical point would force one in every period.) Under (S), T_d is increasing, so $s_j = T_d(j)$ and $\sup_{0 \leq \theta \leq 1} T_d(j+\theta) = T_d(j+1)$. Thus for the intended branch the only genuinely *analytic* inputs remain Hypotheses 2.2(R2) and 2.4 of [1] themselves—the same conditional status as the phase law of [1]; no new type of estimate is introduced by Theorem D.

Remark 2.4 (Local sufficiency). Inspection of the proofs shows that of (S) only the following is actually used: $T_d \in C^1$ with $T'_d > 0$ on $[0, \infty)$, and A_b of class C^1 with $A'_b > 0$ on a compact neighborhood of $D([0, 1])$, together with the Abel identity (5) near $D([0, 1])$ (for Remark 6.8 also near $b^{D([0, 1])}$). The global frame (B1) is assumed for expository convenience and holds for the intended branch anyway.

2.4 Automatic geometric growth of the tower minima

The derivative estimates of Section 6 pivot on the summability $\sum_j 1/s_j < \infty$. Remarkably, this is not an additional hypothesis: the tower functional equation supplies it.

Lemma 2.5 (Automatic geometric growth). *Assume (R1) of Hypothesis 2.2 of [1] and $s_j \rightarrow \infty$ from Hypothesis 2.4 of [1]. Then*

$$s_{j+1} = d^{s_j} \quad \text{for all } j \geq 0,$$

and consequently there is J_1 with $s_{j+1} \geq 2s_j$ for all $j \geq J_1$; in particular

$$\sum_{j \geq 1} \frac{1}{s_j} < \infty.$$

Proof. Fix j . Taking the infimum over $\theta \in [0, 1]$ in $T_d(j+1+\theta) = d^{T_d(j+\theta)}$ and using that $t \mapsto d^t$ is increasing and continuous gives $s_{j+1} = d^{s_j}$; both infima run over the same compact phase interval, and $s_j > 0$ by (R1). Since $d > 1$, $d^t/t \rightarrow \infty$, so there is $t_0 > 0$ with $d^t \geq 2t$ for $t \geq t_0$; choose J_1 with $s_j \geq t_0$ for $j \geq J_1$ (possible as $s_j \rightarrow \infty$). Then $s_{J_1+k} \geq 2^k s_{J_1}$, and $\sum_j 1/s_j$ converges by comparison with a geometric series. $\square$

Remark 2.6. Lemma 2.5 shows that the qualitative divergence $s_j \rightarrow \infty$ self-improves to geometric (indeed tower) growth. This is special to the tetration recursion: in an abstract iterated-logarithm scheme lacking the exact functional equation, divergence of $\sum 1/s_j$ with $\log_b \alpha \neq 0$ would make the derivative products of Section 6 degenerate to 0 or ∞ , so summability is exactly the right condition—and here it is free.

2.5 Endpoint identities

The following sharpening of Lemma 5.8 of [1] will be used at both the C^0 and the C^1 level. Only (H1) and the branch frame are needed.

Lemma 2.7 (Endpoint identities). *Assume (H1). Then*

$$D(1) = b^{D(0)}, \quad \text{and consequently} \quad \Phi_{b,d}(1) = \Phi_{b,d}(0). \quad (8)$$

Proof. Since $L_b^{\circ(n+1)} = L_b \circ L_b^{\circ n}$,

$$D_{n+1}(0) = L_b^{\circ(n+1)}(T_d(n+1)) = L_b(L_b^{\circ n}(T_d(n+1))) = L_b(D_n(1)),$$

i.e. $D_n(1) = b^{D_{n+1}(0)}$ for every n for which the quantities are defined. Letting $n \rightarrow \infty$ and using $D_n \rightarrow D$ uniformly together with continuity of $t \mapsto b^t$ gives $D(1) = b^{D(0)}$. Then, by the global Abel equation (5),

$$\Phi_{b,d}(1) = A_b(D(1)) - 1 = A_b(b^{D(0)}) - 1 = A_b(D(0)) = \Phi_{b,d}(0). \quad \square$$

In particular the 1-periodic extension of $\Phi_{b,d}$ is continuous on $\mathbb{R}$, and $x \mapsto \Phi_{b,d}(\{x\})$ is continuous on all of $\mathbb{R}$, *including at the integers*. This last point is what allows fractional parts to be composed safely below.

3 The topological theorem

Throughout this section all phases are extended 1-periodically to $\mathbb{R}$; thus for a pair (p, q) with package (P) we write $\Phi_{p,q} : \mathbb{R} \rightarrow \mathbb{R}$ (continuous, 1-periodic) and $\tilde{H}_{p,q} = \text{id} + \Phi_{p,q}$, so that $\tilde{H}_{p,q}(x+1) = \tilde{H}_{p,q}(x) + 1$ and $\Phi_{p,q}(x) = \Phi_{p,q}(\{x\})$.

Lemma 3.1 (Exact composition of lifted phase maps). *Let (p, q) , (q, r) , (p, r) be ordered pairs for which the package (P) holds. Then*

$$\tilde{H}_{p,r} = \tilde{H}_{p,q} \circ \tilde{H}_{q,r} \quad \text{on } \mathbb{R}.$$

Proof. Fix $\theta \in [0, 1)$ and let $x = n + \theta$, $n \rightarrow \infty$. By Proposition 6.1 of [1], $F_{p,r} = F_{p,q} \circ F_{q,r}$ exactly. By (P) for (q, r) ,

$$y_n := F_{q,r}(n + \theta) = n + \theta + \Phi_{q,r}(\theta) + \varepsilon_n, \quad \varepsilon_n \rightarrow 0.$$

By (P) for (p, q) , $F_{p,q}(y) = y + \Phi_{p,q}(\{y\}) + o(1)$ as $y \rightarrow +\infty$; applying this along $y_n \rightarrow +\infty$ and using that $u \mapsto \Phi_{p,q}(\{u\}) = \Phi_{p,q}(u)$ is continuous on $\mathbb{R}$ (the periodic extension; Lemma 2.7 for (p, q)), we get

$$F_{p,r}(n + \theta) - (n + \theta) = \Phi_{q,r}(\theta) + \varepsilon_n + \Phi_{p,q}(\theta + \Phi_{q,r}(\theta) + \varepsilon_n) + o(1) \longrightarrow \Phi_{q,r}(\theta) + \Phi_{p,q}(\theta + \Phi_{q,r}(\theta)).$$

The left side converges to $\Phi_{p,r}(\theta)$ by (P) for (p, r) . Hence

$$\Phi_{p,r}(\theta) = \Phi_{q,r}(\theta) + \Phi_{p,q}(\theta + \Phi_{q,r}(\theta)), \quad \theta \in [0, 1), \quad (9)$$

and by periodicity for all $\theta \in \mathbb{R}$. (This is Proposition 6.4 of [1]; the continuity of the periodic extension across integer values of $\theta + \Phi_{q,r}(\theta)$, provided by Lemma 2.7, is what makes the limit of the composed fractional parts legitimate.) Adding θ to (9),

$$\tilde{H}_{p,r}(\theta) = (\theta + \Phi_{q,r}(\theta)) + \Phi_{p,q}(\theta + \Phi_{q,r}(\theta)) = \tilde{H}_{p,q}(\tilde{H}_{q,r}(\theta)). \quad \square$$

Proof of Theorem A. Apply Lemma 3.1 to the triples (d, b, d) and (b, d, b) ; since $\Phi_{d,d} \equiv 0 \equiv \Phi_{b,b}$ (diagonal packages are exact), this gives

$$\tilde{H}_{d,b} \circ \tilde{H}_{b,d} = \tilde{H}_{d,d} = \text{id}, \quad \tilde{H}_{b,d} \circ \tilde{H}_{d,b} = \tilde{H}_{b,b} = \text{id}.$$

Thus $\tilde{H} := \tilde{H}_{b,d}$ is a continuous bijection of $\mathbb{R}$ with continuous inverse $\tilde{H}_{d,b}$. A continuous injection of $\mathbb{R}$ is strictly monotone; since $\tilde{H}(x+1) = \tilde{H}(x) + 1 > \tilde{H}(x)$, it is strictly *increasing*. Because $\tilde{H}$ commutes with $x \mapsto x+1$, it descends to a homeomorphism $H_{b,d}$ of S^1 , orientation-preserving since the lift is increasing of degree one, and $H_{b,d}^{-1} = H_{d,b}$. The groupoid statement is Lemma 3.1. $\square$

Remark 3.2. Theorem A upgrades the “nonlinear phase cocycle” of [1, Prop. 6.4] to literal composition in $\text{Homeo}_+(S^1)$: the family $\{H_{p,q}\}$ is a groupoid of circle homeomorphisms, and (9) is nothing but $H_{p,r} = H_{p,q} \circ H_{q,r}$ read in the lift.

4 Antisymmetry of the mean shifts without smoothness

Proof of Theorem B. Write $\Phi = \Phi_{b,d}$, $\Psi = \Phi_{d,b}$ (both 1-periodic and continuous on $\mathbb{R}$), $\tilde{H} = \tilde{H}_{b,d}$, $\tilde{H}_{d,b} = \tilde{H}^{-1}$ (Theorem A). The inverse case of (9) (triple (d, b, d)) reads

$$\Psi(\tilde{H}(\theta)) = -\Phi(\theta), \quad \theta \in \mathbb{R}. \quad (10)$$

Since $\tilde{H}$ is increasing with $\tilde{H}(\theta+1) = \tilde{H}(\theta) + 1$, the function $\Phi = \tilde{H} - \text{id}$ is continuous and of bounded variation on every interval of length 1, with $\text{Var}_{[t,t+1]} \Phi \leq \text{Var}_{[t,t+1]} \tilde{H} + 1 = (\tilde{H}(t+1) - \tilde{H}(t)) + 1 = 2$.

Let $t_0 := \tilde{H}_{d,b}(0)$, so $\tilde{H}(t_0) = 0$ and $\tilde{H}(t_0+1) = 1$. Since $\tilde{H}$ is a continuous increasing bijection of $[t_0, t_0+1]$ onto $[0, 1]$ and Ψ is continuous, the Riemann–Stieltjes change of variables $u = \tilde{H}(\theta)$ [5, Thm. 7.7] gives, using 1-periodicity of Ψ in the first step and (10) in the third,

$$\mu_{d,b} = \int_0^1 \Psi(u) du = \int_{t_0}^{t_0+1} \Psi(\tilde{H}(\theta)) d\tilde{H}(\theta) = - \int_{t_0}^{t_0+1} \Phi(\theta) d(\theta + \Phi(\theta)) = - \int_{t_0}^{t_0+1} \Phi d\theta - \int_{t_0}^{t_0+1} \Phi d\Phi.$$

All Stieltjes integrals exist because the integrands are continuous and the integrators are of bounded variation, and the splitting uses linearity of $g \mapsto \int f dg$. By periodicity of Φ , $\int_{t_0}^{t_0+1} \Phi d\theta = \mu_{b,d}$. For the last integral, integration by parts for Riemann–Stieltjes integrals [5, Thm. 7.6] with $f = g = \Phi$ (continuous, BV) yields

$$2 \int_{t_0}^{t_0+1} \Phi d\Phi = \Phi(t_0+1)^2 - \Phi(t_0)^2 = 0$$

by periodicity. Hence $\mu_{d,b} = -\mu_{b,d}$. $\square$

Remark 4.1. The proof of Proposition 6.6 in [1] is exactly this computation with $d\tilde{H}(\theta) = H'(\theta) d\theta$; the Stieltjes formulation shows that only monotonicity—automatic by Theorem A—is needed, not differentiability. Together with Theorem A, this *de-conditionalizes* Proposition 6.6 of [1] completely.

Remark 4.2 (The smooth substitution, with the interval shift). If in addition $\Phi, \Psi \in C^1$ (as under the hypotheses of Theorem D for both pairs, or of Theorem C), the substitution reads classically:

$$\mu_{d,b} = \int_0^1 \Psi(u) du = \int_{\tilde{H}(0)}^{\tilde{H}(0)+1} \Psi(u) du = \int_0^1 \Psi(\tilde{H}(\theta)) \tilde{H}'(\theta) d\theta = - \int_0^1 \Phi(1+\Phi') d\theta = -\mu_{b,d} - \frac{1}{2} [\Phi^2]_0^1 = -\mu_{b,d},$$

where the first equality uses 1-periodicity of Ψ : the map $\tilde{H}$ carries $[0, 1]$ onto $[\tilde{H}(0), \tilde{H}(0) + 1]$ rather than onto $[0, 1]$, and periodicity supplies the missing shift. This is the computation of [1, Prop. 6.6] with that point made explicit.

5 A minimal-regularity principle

Theorem C is a purely formal statement about pairs of 1-periodic C^1 functions coupled by the inverse phase equations; it consumes no channel estimates and no derivative formulas.

Proof of Theorem C. The equations (2) give, for every $x \in \mathbb{R}$,

$$\tilde{K}(\tilde{H}(x)) = \tilde{H}(x) + \Psi(\tilde{H}(x)) = x + \Phi(x) - \Phi(x) = x,$$

and symmetrically $\tilde{H}(\tilde{K}(u)) = u$; thus $\tilde{K} \circ \tilde{H} = \tilde{H} \circ \tilde{K} = \text{id}$ with $\tilde{H}, \tilde{K} \in C^1(\mathbb{R})$. Differentiating the first identity by the chain rule,

$$\tilde{K}'(\tilde{H}(x)) \tilde{H}'(x) = 1 \quad \text{for all } x \in \mathbb{R},$$

so $\tilde{H}'$ has no zeros. Being continuous on the connected set $\mathbb{R}$, $\tilde{H}'$ has a constant sign, and

$$\int_0^1 \tilde{H}'(x) dx = \tilde{H}(1) - \tilde{H}(0) = 1 > 0$$

(1-periodicity of Φ) forces the sign to be positive: $\tilde{H}' > 0$ everywhere, and $\tilde{K}' = 1/(\tilde{H}' \circ \tilde{K}) > 0$. A C^1 map of $\mathbb{R}$ with everywhere positive derivative and $\tilde{H}(x+1) = \tilde{H}(x) + 1$ is a strictly increasing bijection of $\mathbb{R}$; its inverse $\tilde{K}$ is C^1 , both commute with the unit translation, and so they descend to mutually inverse orientation-preserving C^1 diffeomorphisms of S^1 . Finally, under (H1)+(H2) the equations (2) for $\Phi = \Phi_{b,d}$, $\Psi = \Phi_{d,b}$ are the cases (d, b, d) and (b, d, b) of Lemma 3.1. $\square$

Example 5.1 (One-sided C^1 regularity does not suffice). Let

$$\Phi_0(x) := -\frac{\sin(2\pi x)}{2\pi}, \quad \tilde{H}_0 := \text{id} + \Phi_0.$$

Then Φ_0 is 1-periodic and real-analytic, $\tilde{H}_0'(x) = 1 - \cos(2\pi x) \geq 0$ vanishes exactly at $x \in \mathbb{Z}$, and $\tilde{H}_0$ is strictly increasing: for $y > x$, $\tilde{H}_0(y) - \tilde{H}_0(x) = \int_x^y (1 - \cos 2\pi t) dt > 0$ since the integrand vanishes only at isolated points. Hence $\tilde{H}_0$ is the lift of a degree-one C^1 circle *homeomorphism* which is *not* a C^1 diffeomorphism: near $x = 0$,

$$\tilde{H}_0(x) = \frac{2\pi^2}{3} x^3 + O(x^5),$$

so the inverse behaves like a cube root at 0 and is not differentiable there; the inverse phase $\Psi_0 = \tilde{H}_0^{-1} - \text{id}$ is continuous but not C^1 . Consequently, C^1 regularity plus topological invertibility of the *forward* lift alone do not imply the diffeomorphism property. Theorem C genuinely consumes C^1 regularity of both phases, while Theorem D supplies the positivity $\tilde{H}' > 0$ for the forward pair outright.

Corollary 5.2. *Assume (H1) and (H2). If $\Phi_{b,d}$ and $\Phi_{d,b}$ are of class C^1 on the circle—however this is established—then $H_{b,d}$ is an orientation-preserving C^1 circle diffeomorphism with C^1 inverse $H_{d,b}$. In particular the separate diffeomorphism assumption in [1, Prop. 6.6] is redundant given two-sided C^1 phase limits.*

Remark 5.3 (Logical role). Theorem C decouples the diffeomorphism property from the specific analytic mechanism: any route to two-sided C^1 phases inherits the full conclusion. For instance, a proof of strip analyticity ([1, Conj. 8.1]; Remark 7.3) for both ordered pairs would immediately produce mutually inverse real-analytic circle diffeomorphisms. Within the present note it is not strictly needed, since Theorem D(i) proves $\tilde{H}' > 0$ directly.

6 The analytic C^1 theorem

Standing assumptions for this section: (H1), (S), (R'), together with the branch frame (B1)–(B2). We abbreviate $W_k = W_k^{(n)}(\theta) = L_b^{\circ k}(T_d(n + \theta))$, so $W_0 = T_d(n + \theta)$, $W_{k+1} = L_b(W_k)$, $W_n = D_n(\theta)$, and, by (R'), $W_k > 0$ for $0 \leq k \leq n-1$ and $n \geq N_0$. Since $W_{k+1} = L_b(W_k)$ with $W_k > 0$ is equivalent to $W_k = b^{W_{k+1}}$, we record the inversion

$$W_{n-j}^{(n)}(\theta) = \exp_b^{\circ j}(D_n(\theta)), \quad 1 \leq j \leq n. \quad (11)$$

6.1 The exact derivative product identity

Lemma 6.1 (Exact product identity). *Let $n \geq \max(N_0, 2)$ and $\theta \in [0, 1]$. Then $D_n \in C^1([0, 1])$, and with*

$$1 + \sigma_j^{(n)}(\theta) := \frac{W_{n-j}^{(n)}(\theta)}{\alpha T_d(j + \theta)} = \frac{\exp_b^{\circ j}(D_n(\theta))}{\alpha T_d(j + \theta)} > 0, \quad 1 \leq j \leq n-1, \quad (12)$$

one has the exact identity

$$D'_n(\theta) = \frac{\alpha T'_d(\theta)}{\prod_{j=1}^{n-1} (1 + \sigma_j^{(n)}(\theta))}, \quad \text{and} \quad \sigma_{n-1}^{(n)} \equiv 0. \quad (13)$$

Proof. By (S) and (R'), each map in the chain $\theta \mapsto T_d(n + \theta) = W_0 \mapsto W_1 \mapsto \cdots \mapsto W_n = D_n(\theta)$ is C^1 along the orbit (the logarithms are applied to positive quantities, which range over a compact subset of $(0, \infty)$ as $\theta \in [0, 1]$), so $D_n \in C^1([0, 1])$ and the chain rule with $L'_b(y) = 1/(y \ln b)$ gives

$$D'_n(\theta) = \frac{T'_d(n + \theta)}{(\ln b)^n \prod_{k=0}^{n-1} W_k}. \quad (14)$$

Differentiating $T_d(x + 1) = d^{T_d(x)}$ yields $T'_d(x + 1) = (\ln d)T_d(x + 1)T'_d(x)$, and iterating from $x = \theta$,

$$T'_d(n + \theta) = (\ln d)^n T'_d(\theta) \prod_{i=1}^n T_d(i + \theta). \quad (15)$$

Next, the first peel is exact:

$$W_1 = L_b(T_d(n + \theta)) = L_b(d^{T_d(n-1+\theta)}) = \alpha T_d(n - 1 + \theta),$$

which is (12) with $j = n - 1$ and $\sigma_{n-1}^{(n)} = 0$. Using (12) for $1 \leq k \leq n - 1$ (i.e. $j = n - k$),

$$\prod_{k=0}^{n-1} W_k = T_d(n + \theta) \cdot \alpha^{n-1} \prod_{i=1}^{n-1} T_d(i + \theta) \cdot \prod_{j=1}^{n-1} (1 + \sigma_j^{(n)}(\theta)).$$

Substituting this and (15) into (14), the factor $(\ln d)^n / (\ln b)^n = \alpha^n$ appears, the products $\prod_{i=1}^n T_d(i + \theta)$ and $T_d(n + \theta) \prod_{i=1}^{n-1} T_d(i + \theta)$ cancel exactly, and $\alpha^n / \alpha^{n-1} = \alpha$ survives:

$$D'_n(\theta) = \frac{\alpha^n T'_d(\theta) \prod_{i=1}^n T_d(i + \theta)}{\alpha^{n-1} T_d(n + \theta) \prod_{i=1}^{n-1} T_d(i + \theta) \prod_{j=1}^{n-1} (1 + \sigma_j^{(n)})} = \frac{\alpha T'_d(\theta)}{\prod_{j=1}^{n-1} (1 + \sigma_j^{(n)})}.$$

The identification (12) of W_{n-j} with $\exp_b^{\circ j}(D_n)$ is (11). $\square$

Note that (13) is an algebraic identity: it holds for *any* C^1 germ propagated by the tower recursion, independently of the branch choice; see Remark 7.5 for a high-precision numerical confirmation.

Remark 6.2 (Ratio identity and an alternative route). Dividing the chain-rule products for consecutive tails yields a companion identity. By [1, eq. (22)], $D_{n+1}(\theta) = L_b^{\circ n}(\alpha T_d(n + \theta))$; writing $\tilde{V}_k := L_b^{\circ k}(\alpha T_d(n + \theta))$ and $\delta_k := \tilde{V}_k - W_k$ —the damped differences of [1, Lemma 5.3], with $\delta_1 = L_b(\alpha)$ —one obtains, since $\tilde{V}_0 = \alpha W_0$ and all quantities are positive by (R') at level $n + 1$,

$$\frac{D'_{n+1}(\theta)}{D'_n(\theta)} = \prod_{k=1}^{n-1} \frac{W_k}{W_k + \delta_k}.$$

The huge factor $T'_d(n + \theta)$ cancels identically. With the estimates of [1, Lemma 5.3] one gets $\|\log D'_{n+1} - \log D'_n\|_\infty = O(1/s_{n-1})$, summable in n by Lemma 2.5; hence $(\log D'_n)$ is uniformly Cauchy, giving an alternative proof of Proposition 6.6 that inherits the hyper-exponential speed of [1, Lemma 5.4]. We follow the closed-form route below because it produces the limit formula (3) and the positivity bound in the same stroke.

6.2 Tail estimates

Lemma 6.3 (Recursion and tail bound). *For $n \geq \max(N_0, 2)$, $\theta \in [0, 1]$, and $1 \leq j \leq n - 2$,*

$$\sigma_j^{(n)}(\theta) = \frac{L_b(\alpha) + L_b(1 + \sigma_{j+1}^{(n)}(\theta))}{\alpha T_d(j + \theta)}, \quad \sigma_{n-1}^{(n)} = 0. \quad (16)$$

Let

$$C_0 := \frac{|L_b(\alpha)| + \frac{\ln 2}{\ln b}}{\alpha}, \quad J_0 := \min \left\{ J : \frac{C_0}{s_j} \leq \frac{1}{2} \text{ for all } j \geq J \right\}$$

(finite since $s_j \rightarrow \infty$). Then for all $n \geq \max(N_0, 2)$ and $\theta \in [0, 1]$,

$$|\sigma_j^{(n)}(\theta)| \leq \frac{C_0}{s_j} \leq \frac{1}{2} \quad \text{for } J_0 \leq j \leq n - 1. \quad (17)$$

Proof. For $1 \leq j \leq n-2$, using (12) at level $j+1$, the identity $L_b(T_d(j+1+\theta)) = L_b(d^{T_d(j+\theta)}) = \alpha T_d(j+\theta)$, and additivity of L_b on positive factors,

$$W_{n-j} = L_b(W_{n-j-1}) = L_b\left(\alpha T_d(j+1+\theta)(1+\sigma_{j+1}^{(n)})\right) = L_b(\alpha) + \alpha T_d(j+\theta) + L_b(1+\sigma_{j+1}^{(n)});$$

dividing by $\alpha T_d(j+\theta)$ gives (16). For (17) we induct downward on j from $j = n-1$ to $j = J_0$. The base case is $\sigma_{n-1}^{(n)} = 0$. If $|\sigma_{j+1}^{(n)}| \leq \frac{1}{2}$ (with $J_0 \leq j \leq n-2$), then $|\ln(1+\sigma_{j+1}^{(n)})| \leq \ln 2$, so (16) gives

$$|\sigma_j^{(n)}| \leq \frac{|L_b(\alpha)| + \frac{\ln 2}{\ln b}}{\alpha T_d(j+\theta)} \leq \frac{C_0}{s_j} \leq \frac{1}{2}. \quad \square$$

Lemma 6.4 (Head control and limit factors). *There is $N_1 \geq \max(N_0, 2)$ such that for every fixed $j \geq 1$:*

(a) $1 + \sigma_j^{(n)} \rightarrow 1 + \sigma_j$ uniformly on $[0, 1]$ as $n \rightarrow \infty$, where

$$1 + \sigma_j(\theta) := \frac{E_j(\theta)}{\alpha T_d(j+\theta)}, \quad E_j(\theta) := \exp_b^{\circ j}(D(\theta)) > 0;$$

(b) *there are constants $0 < m_j \leq M_j < \infty$ with $m_j \leq 1 + \sigma_j^{(n)}(\theta) \leq M_j$ for all $n \geq N_1$, $\theta \in [0, 1]$, and $m_j \leq 1 + \sigma_j(\theta) \leq M_j$;*

(c) $|\sigma_j(\theta)| \leq C_0/s_j$ for $j \geq J_0$.

Proof. By (H1), $D_n \rightarrow D$ uniformly on $[0, 1]$; choose $N_1 \geq \max(N_0, 2)$ with $\|D_n - D\|_\infty \leq 1$ for $n \geq N_1$ and set $K := [\min_{[0,1]} D - 1, \max_{[0,1]} D + 1]$, a compact interval containing all values $D_n(\theta)$ ($n \geq N_1$) and $D(\theta)$. Each $\exp_b^{\circ j}$ is C^1 , hence Lipschitz on K , and $\exp_b^{\circ j}(K) =: [c_j, C_j] \subset (0, \infty)$ for $j \geq 1$. By (11), $W_{n-j}^{(n)} = \exp_b^{\circ j}(D_n) \rightarrow \exp_b^{\circ j}(D) = E_j$ uniformly, which gives (a); moreover $E_j \geq c_j > 0$. By (B2) and continuity, $T_d(j+\cdot)$ has a finite supremum S_j and infimum $s_j > 0$ on $[0, 1]$, so (b) holds with $m_j = c_j/(\alpha S_j)$, $M_j = C_j/(\alpha s_j)$. Passing to the limit $n \rightarrow \infty$ in (17) gives (c). $\square$

6.3 Uniform convergence of the product and C^1 convergence

Lemma 6.5 (Uniform product convergence). *Set*

$$\Pi_n(\theta) := \prod_{j=1}^{n-1} (1 + \sigma_j^{(n)}(\theta)), \quad \Pi(\theta) := \prod_{j=1}^{\infty} (1 + \sigma_j(\theta)).$$

Then Π converges absolutely and uniformly on $[0, 1]$, is continuous, satisfies $0 < c_\Pi \leq \Pi \leq C_\Pi < \infty$, and $\Pi_n \rightarrow \Pi$ uniformly on $[0, 1]$.

Proof. All factors are positive (Lemma 6.1, Lemma 6.4(b)), so we may work with logarithms. For $|s| \leq \frac{1}{2}$ one has $|\ln(1+s)| \leq 2|s|$; hence by (17) and Lemma 6.4(c),

$$|\ln(1 + \sigma_j^{(n)})| \leq \frac{2C_0}{s_j}, \quad |\ln(1 + \sigma_j)| \leq \frac{2C_0}{s_j} \quad (j \geq J_0), \quad (18)$$

uniformly in n and θ . By Lemma 2.5, $\sum_j 2C_0/s_j < \infty$, which already gives absolute and uniform convergence of $\ln \Pi = \sum_{j \geq 1} \ln(1 + \sigma_j)$ (the finitely many terms $j < J_0$ are bounded by Lemma 6.4(b)), hence continuity of Π and the two-sided bound with $\ln C_\Pi = -\ln c_\Pi := \sum_{j < J_0} \max(|\ln m_j|, |\ln M_j|) + \sum_{j \geq J_0} 2C_0/s_j$.

For $\Pi_n \rightarrow \Pi$: given $\varepsilon > 0$, choose $J \geq J_0$ with $\sum_{j \geq J} 2C_0/s_j < \varepsilon/3$ (Lemma 2.5). For $n - 1 \geq J$,

$$|\ln \Pi_n - \ln \Pi| \leq \sum_{j=1}^{J-1} \left| \ln(1 + \sigma_j^{(n)}) - \ln(1 + \sigma_j) \right| + \sum_{j=J}^{n-1} |\ln(1 + \sigma_j^{(n)})| + \sum_{j=J}^{\infty} |\ln(1 + \sigma_j)|.$$

The two tails are each $< \varepsilon/3$ by (18). In the head, each of the finitely many terms tends to 0 uniformly by Lemma 6.4(a) and the Lipschitz continuity of $\ln$ on $[m_j \wedge \inf(1 + \sigma_j), \infty) \subset [m_j, \infty)$. Hence $\limsup_n \|\ln \Pi_n - \ln \Pi\|_\infty \leq \frac{2\varepsilon}{3} < \varepsilon$, so $\ln \Pi_n \rightarrow \ln \Pi$ uniformly, and applying $\exp$ (uniformly continuous on the compact range) gives $\Pi_n \rightarrow \Pi$ uniformly. $\square$

Proposition 6.6 (C^1 convergence of the tails). $D_n \rightarrow D$ in $C^1([0, 1])$; $D \in C^1([0, 1])$; and

$$D'(\theta) = \frac{\alpha T'_d(\theta)}{\Pi(\theta)} = \alpha T'_d(\theta) \prod_{j=1}^{\infty} \frac{\alpha T_d(j + \theta)}{E_j(\theta)} \geq \frac{\alpha \min_{[0,1]} T'_d}{C_\Pi} > 0.$$

Proof. By Lemma 6.1 and Lemma 6.5, $D'_n = \alpha T'_d/\Pi_n \rightarrow \alpha T'_d/\Pi =: G$ uniformly on $[0, 1]$, with G continuous and, by (S) and Lemma 6.5, bounded below by the stated positive constant. Since also $D_n \rightarrow D$ uniformly (H1), the standard theorem on differentiation of uniform limits [6, Thm. 7.17] yields $D \in C^1$ and $D' = G$. The product form of G is (12)–(13) in the limit, i.e. $(1 + \sigma_j)^{-1} = \alpha T_d(j + \theta)/E_j$. $\square$

6.4 Seam matching

Lemma 6.7 (C^1 seam identities). *Under the standing assumptions,*

$$D(1) = b^{D(0)}, \quad D'(1) = (\ln b) D(1) D'(0),$$

and consequently, with $\Phi = \Phi_{b,d} = A_b \circ D - \text{id}$ on $[0, 1]$,

$$\Phi(1) = \Phi(0), \quad \Phi'(1) = \Phi'(0).$$

Proof. The value identities are Lemma 2.7. For the derivative identity, evaluate the product formula of Proposition 6.6 at $\theta = 1$ and $\theta = 0$. First, $T'_d(1) = (\ln d) T_d(1) T'_d(0)$ by (7). Second, by $D(1) = b^{D(0)}$,

$$E_j(1) = \exp_b^{\circ j}(D(1)) = \exp_b^{\circ j}(b^{D(0)}) = \exp_b^{\circ(j+1)}(D(0)) = E_{j+1}(0), \quad j \geq 1.$$

Hence, shifting the product index ($i = j + 1$),

$$D'(1) = \alpha (\ln d) T_d(1) T'_d(0) \prod_{j=1}^{\infty} \frac{\alpha T_d(j+1)}{E_{j+1}(0)} = \alpha (\ln d) T_d(1) T'_d(0) \prod_{i=2}^{\infty} \frac{\alpha T_d(i)}{E_i(0)},$$

while $D'(0) = \alpha T'_d(0) \prod_{i=1}^{\infty} \alpha T_d(i)/E_i(0)$. All products converge to nonzero limits (Lemma 6.5), so the termwise quotient is legitimate:

$$\frac{D'(1)}{D'(0)} = (\ln d) T_d(1) \cdot \left(\frac{\alpha T_d(1)}{E_1(0)} \right)^{-1} = \frac{\ln d}{\alpha} E_1(0) = (\ln b) b^{D(0)} = (\ln b) D(1).$$

Now differentiate the global Abel equation (5) (both sides C^1 by (S)) at $y = D(0)$:

$$A'_b(b^y) b^y \ln b = A'_b(y) \implies A'_b(D(1)) (\ln b) D(1) = A'_b(D(0)).$$

Therefore

$$\Phi'(1) + 1 = A'_b(D(1)) D'(1) = \frac{A'_b(D(0))}{(\ln b) D(1)} \cdot (\ln b) D(1) D'(0) = A'_b(D(0)) D'(0) = \Phi'(0) + 1. \quad \square$$

Remark 6.8 (Seam via extended phases). Alternatively: the exact transport $D_n(1+\theta) = b^{D_{n+1}(\theta)}$ (the inversion (11) one level up; positivity by (R') at level $n+1$) carries the uniform convergence of D_{n+1} on $[0, 1]$ to that of D_n on $[1, 2]$, and every estimate of this section holds verbatim with phase range $[0, 2]$ (with s_j replaced by $\min(s_j, s_{j+1})$ in the tail bounds). Running Proposition 6.6 on $[0, 2]$ gives $D \in C^1([0, 2])$ with $D(1+\theta) = b^{D(\theta)}$, whence $\Phi(1+\theta) = \Phi(\theta)$ holds as an identity of C^1 functions on $[0, 1]$ —including the derivative match at the seam—without touching the product formula. This formulation extends verbatim to higher derivatives; see Remark 7.3.

6.5 Proof of Theorem D and corollaries

Proof of Theorem D. (i) is Proposition 6.6.

(ii) By (S) and (6), $A_b \in C^1(\mathbb{R})$ with $A'_b > 0$, so $\Phi = A_b \circ D - \text{id} \in C^1([0, 1])$ with $\Phi' = (A'_b \circ D) D' - 1$. By Lemma 6.7, Φ has matching values and one-sided derivatives at the endpoints, so its 1-periodic extension is C^1 on $\mathbb{R}$ (the extension is differentiable at the integers with derivative $\Phi'(0) = \Phi'(1)$, and Φ' extends continuously and periodically). Hence $\tilde{H} = \tilde{H}_{b,d} = \text{id} + \Phi \in C^1(\mathbb{R})$, $\tilde{H}(x+1) = \tilde{H}(x) + 1$, and on $[0, 1]$ (hence on $\mathbb{R}$ by periodicity of $\tilde{H}'$)

$$\tilde{H}' = 1 + \Phi' = (A'_b \circ D) D' \geq \delta_0 := \min_{[0,1]} [(A'_b \circ D) D'] > 0,$$

using Proposition 6.6 and continuity on the compact interval. For the closed form (4): since $\tilde{H}(\theta) = A_b(D(\theta))$, we have $T_b(\tilde{H}(\theta)) = D(\theta)$, hence by iterating $T_b(x+1) = b^{T_b(x)}$,

$$T_b(j + \tilde{H}(\theta)) = \exp_b^{\circ j} (T_b(\tilde{H}(\theta))) = \exp_b^{\circ j} (D(\theta)) = E_j(\theta), \quad j \geq 1,$$

and $A'_b(D(\theta)) = 1/T'_b(A_b(D(\theta))) = 1/T'_b(\tilde{H}(\theta))$ by (6); substituting into $\tilde{H}' = (A'_b \circ D) D'$ and (3) gives (4).

(iii) $\tilde{H}$ is C^1 with $\tilde{H}' \geq \delta_0 > 0$ and $\tilde{H}(x+1) = \tilde{H}(x) + 1$; hence $\tilde{H}$ is a strictly increasing C^1 bijection of $\mathbb{R}$ (surjectivity from the translation property), its inverse is C^1 by the inverse function theorem, and both $\tilde{H}$ and $\tilde{H}^{-1}$ commute with the unit translation. Therefore $\tilde{H}$ descends to an orientation-preserving C^1 diffeomorphism $H_{b,d}$ of S^1 . This is precisely the hypothesis of Proposition 6.6 of [1]. If moreover (H2) holds, Theorem A identifies $\tilde{H}^{-1} = \tilde{H}_{d,b} = \text{id} + \Phi_{d,b}$, so $\Phi_{d,b} = \tilde{H}^{-1} - \text{id} \in C^1$, i.e. the reversed phase is automatically C^1 on the circle. (Alternatively, once both phases are known to be C^1 , Theorem C applies.)

(iv) By Proposition 5.1 of [1], $R_n(\theta) = F_{b,d}(n+\theta) - (n+\theta) = A_b(D_n(\theta)) - \theta$ exactly. Hence $R_n \in C^1([0, 1])$ and $R'_n = (A'_b \circ D_n) D'_n - 1$. Since $D_n \rightarrow D$ uniformly with values in the compact K (Lemma 6.4), A'_b is uniformly continuous on K , and $D'_n \rightarrow D'$ uniformly (Proposition 6.6), we get $R'_n \rightarrow (A'_b \circ D) D' - 1 = \Phi'$ uniformly; together with $R_n \rightarrow \Phi$ uniformly (Theorem 5.7 of [1]) this is C^1 convergence. Finally $F'_{b,d}(n+\theta) = 1 + R'_n(\theta) \rightarrow 1 + \Phi'(\theta) = \tilde{H}'(\theta)$. $\square$

Corollary 6.9 (C^1 groupoid). *If the hypotheses of Theorem D hold for all ordered pairs from $\{b, c, d\}$, then all maps $H_{p,q}$ are orientation-preserving C^1 circle diffeomorphisms and $H_{p,r} = H_{p,q} \circ H_{q,r}$ in $\text{Diff}_+^1(S^1)$; the identity $\Phi_{p,r} = \Phi_{q,r} + \Phi_{p,q} \circ (\text{id} + \Phi_{q,r})$ of [1, Prop. 6.4] is its expression in lifts.*

Proof. Combine Theorem D with Lemma 3.1. □

Corollary 6.10 (Proposition 6.6 of [1], discharged). *Under (H1) and (H2): $\mu_{b,d} = -\mu_{d,b}$ (Theorem B; no smoothness). Under (H1), (S), (R'): $H_{b,d}$ is an orientation-preserving C^1 circle diffeomorphism, i.e. the literal hypothesis of [1, Prop. 6.6] holds; if (H2) holds as well, so does its inverse-side counterpart, with $H_{d,b} = H_{b,d}^{-1}$ of class C^1 .*

Proof. Immediate from Theorems B and D. □

7 Remarks

Remark 7.1 (The mismatch sits one level down). By (12) in the limit and Lemma 6.4(c),

$$T_b(j + \tilde{H}_{b,d}(\theta)) = E_j(\theta) = \alpha T_d(j + \theta) (1 + \sigma_j(\theta)), \quad |\sigma_j(\theta)| \leq \frac{C_0}{s_j} \quad (j \geq J_0),$$

uniformly in θ . Thus measuring the d -tower at height $j + \theta$ in b -coordinates at the *phase-corrected* height $j + \tilde{H}_{b,d}(\theta)$ reproduces the value up to the single factor α and a hyper-exponentially small relative error: the base mismatch survives exactly one tower level down. This quantifies, at the level of values, the damping mechanism behind Theorem 5.6 of [1], and it is the reason the derivative product (4) converges so violently fast.

Remark 7.2 (Factorization $D = T_b \circ \tilde{H}$). Since $\tilde{H}(\theta) = A_b(D(\theta))$, one has $D = T_b \circ \tilde{H}$ on $[0, 1]$: the limiting tail is the phase map read in the T_b -coordinate. With the extension of Remark 6.8, the translation identity $D(1 + \theta) = b^{D(\theta)}$ is precisely the tetration equation of T_b transported by $\tilde{H}$. Given the branch regularity (S), regularity of D and regularity of $\tilde{H}$ are therefore equivalent.

Remark 7.3 (C^r and analytic upgrades). If in (S) the branches are of class C^r ($r \geq 2$), the same scheme yields $\Phi_{b,d} \in C^r(S^1)$ and a C^r diffeomorphism: differentiate $\log D'_n = \log(\alpha T'_d) - \sum_j \log(1 + \sigma_j^{(n)})$ repeatedly, or run the extension bookkeeping of Remark 6.8; the tail control is again Lemma 6.3 together with Lemma 2.5. The proof is likewise stable under complexification: if the channel estimates and the uniform convergence $D_n \rightarrow D$ hold on a strip $\{|\text{Im } \theta| < \sigma_0\}$ (with T_d, T_b holomorphic there, as they are for the regular branch), Weierstrass' theorem upgrades Proposition 6.6 to locally uniform convergence of holomorphic functions; then $D, \Phi_{b,d}$ and $\tilde{H}_{b,d}$ are real-analytic, the Fourier coefficients of $P_{b,d}$ decay exponentially, and Conjecture 8.1 of [1] follows—by Theorem C, both phase maps are then automatically mutually inverse *analytic* circle diffeomorphisms. The product identity (13) is already the correct analytic object for that program; we do not pursue the complex channel estimates here.

Remark 7.4 (What remains conditional). Theorems A and B consume only the conclusions of the phase theorem of [1] (for both ordered pairs); Theorem C is unconditional as a statement about C^1 data; and Theorem D consumes, beyond (H1), only (S) (true for the regular branch, and reducible to one period by Lemma 2.2) and (R') (the quantitative reading of the paper's (R2)); the growth input is supplied by Lemma 2.5. Thus the results are conditional in exactly

the same sense as the main theorem of [1]: everything reduces to the real-channel and peeling-growth estimates for the regular branch (Section 8.3 of [1]); no additional analytic difficulty is introduced by the diffeomorphism statement.

Remark 7.5 (Numerical validation). (i) *The exact identity.* Identity (13) holds for any C^1 germ propagated by the tower recursion, so it can be tested independently of any branch construction: take an arbitrary C^1 function $f > 0$ near θ_0 , define $T_0 = f$, $T_j = d^{T_{j-1}}$, $D_n = L_b^{o_n}(T_n)$, and compare $D'_n(\theta_0)$ (central finite differences, step 10^{-30} , 60-digit arithmetic) with $\alpha f'(\theta_0) \prod_{j=1}^{n-1} \alpha T_j(\theta_0) / \exp_b^{oj}(D_n(\theta_0))$. For $(b, d) = (e, 2)$ at $n = 4, 5$ and $(b, d) = (2, 10)$ at $n = 3, 4$ (seeds $1 + \theta + 0.3 \sin(2.7\theta + 0.4)$ and $0.2(1 + \theta) + 0.05 \sin(2.7\theta + 0.4)$, $\theta_0 = 0.37$), the two sides agree to 31–33 significant digits—the finite-difference accuracy limit—and the inversion (11) holds to working precision ($< 10^{-58}$ relative).

(ii) *End-to-end validation on explicit C^1 branches.* For a base $a > 1$ let

$$q_a(x) := 1 + c_1 x + c_2 x^2 \quad \text{on } [-1, 0], \quad c_1 = \frac{2 \ln a}{1 + \ln a}, \quad c_2 = c_1 - 1,$$

extended to $(-2, \infty)$ by the tetration equation. Then T_a is a genuine C^1 solution of the frame (B1)–(B2) with $T'_a > 0$: the C^1 match at the integers is exactly the constraint defining c_1 , one has $q'_a \geq \min(c_1, 2 - c_1) > 0$ on $[-1, 0]$, and Lemma 2.2 propagates; for $a = e$ this is the classical construction with linear critical piece, $T_e(\theta) = e^\theta$ on $[0, 1]$. On these branches all hypotheses of Theorem D are directly verifiable, and the recursion (16) doubles as the numerically stable evaluation scheme—the interleaved peeling of [1, App. A] in σ -coordinates: truncating $\sigma_{j^*} = 0$ at the first level with $T_d(j^* + \theta)$ beyond working precision reproduces the finite- n objects exactly and the limits to full precision. At 60-digit precision, for the pairs $(e, 2)$ and $(2, e)$: the σ -scheme agrees with direct tower-plus-peeling evaluation to below 10^{-60} ; the increments $\|\Phi_{n+1} - \Phi_n\|$ at $\theta = 0.37$ fall as $2.2 \cdot 10^{-3}$ ($n=4 \rightarrow 5$) and $4.6 \cdot 10^{-19}$ ($n=5 \rightarrow 6$) for $(e, 2)$, and as $8.5 \cdot 10^{-4}$ ($n=3 \rightarrow 4$) and $2.4 \cdot 10^{-34}$ ($n=4 \rightarrow 5$) for $(2, e)$, exhibiting the predicted hyper-exponential collapse; the closed formula (4) matches central finite differences at the finite-difference accuracy limit ($\approx 10^{-36}$); the seam residuals $\Phi(1) - \Phi(0)$ and $\tilde{H}'(1) - \tilde{H}'(0)$ vanish to working precision; $\min_\theta \tilde{H}' \approx 0.9509$ for $(e, 2)$ and ≈ 0.9384 for $(2, e)$, with $\int_0^1 \tilde{H}' = 1$ up to quadrature error; the inverse identity $\tilde{H}_{2,e} \circ \tilde{H}_{e,2} = \text{id}$ holds pointwise to below 10^{-60} ; and

$$\mu_{e,2} = -1.129041157430\dots, \quad \mu_{2,e} = +1.129041157430\dots, \quad \mu_{e,2} + \mu_{2,e} \approx 3 \cdot 10^{-14}$$

at 4096 phase samples (quadrature-limited), confirming Theorem B exactly on a branch different from the regular one. (Incidentally, these toy mean shifts differ from the regular-branch values $\mp 1.128403\dots$ of [1, Table 1] only in the third decimal.)

Authorship and AI Contribution Statement

This paper is part of a five-paper series produced in a directed human–AI collaboration. The author conceived the mixed-base phase program, built and maintains the high-precision tetration engine used for every computation in the series (a fork of Levenstein’s `fatou.gp`; engine, data and scripts at <https://github.com/Lightrunnerwastaken/gp-tetration>), and directed the research: posing the questions, setting priorities between rounds, auditing results, and deciding which claims met the acceptance gates. Substantial parts of the mathematical development and of the numerical work were produced by AI systems under that direction: Anthropic’s Claude

in a theory and review role, an autonomous research agent operating the numerical pipeline, and OpenAI’s ChatGPT, which contributed to the original versions of Papers I and II and to manuscript review. In line with prevailing journal policies, AI systems are not listed as authors; the author assumes full and sole responsibility for all statements and proofs, and readers should weigh this provenance in their assessment.

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